

Raj Tiwari

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (Cyber Security) – CGPA: 8.68

Aug 2023 – Ongoing

Bhopal, India

TECHNICAL PROJECTS

RAGForge | Automated RAG Optimization & Benchmarking Platform | [GitHub](#)

- Built **RAGForge**, a full-stack **RAG optimization platform** (FastAPI, PostgreSQL, ChromaDB, Streamlit, Docker, Gemini).
- Benchmarked **4 tunable parameters** — chunk size, chunk overlap, top-K retrieval, and embedding model — replacing manual trial-and-error tuning.
- Engineered a benchmarking engine spanning **2+ embedding models** (MiniLM, BGE) across **10+ pipeline configurations** per run with ranked leaderboard analytics.
- Delivered a complete **RAG pipeline** (ingestion, chunking, embedding, retrieval, generation) with results persisted in **PostgreSQL** for cross-run comparison.

VidSense AI | Python, FastAPI, LangChain, FAISS, Hugging Face | [GitHub](#)

- Architected a **RAG system** for real-time Q&A over YouTube videos using **FAISS** and **Hugging Face LLMs**.
- Integrated **transcript ingestion, semantic chunking, and dense vector embeddings, cutting hallucinations by 40–50%** vs. naive prompting.
- Designed a **memory-aware, multi-turn** reasoning pipeline with **query rewriting**, sustaining **sub-second latency** and **>90% precision** across **100+ queries**.
- Ran **RAGAS** evaluation (**Context Relevance ≈ 1.0 , Answer Relevancy $\approx 0.75+$**) across a **3+ LLM** abstraction layer.

EXPERIENCE & ACHIEVEMENTS

Eklavya Solution - Remote

May 2026 – Jun 2026

Artificial Intelligence Intern

- Developed and tested **AI-powered application** features in **Python**, resolving **15+ bugs** under senior developer guidance.
- Implemented **5+ REST API** integrations and automated **data preprocessing** for **10+** datasets to support AI components.
- Conducted **15+ code reviews** and maintained **Git version control** across a **5-member** development team.
- Performed **software testing** that reduced **pre-release defects by 20%** prior to deployment.

TECHNICAL SKILLS

Programming Languages: Python, SQL, CPP

AI & LLM Engineering: NLP, Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Evaluation, Vector Databases, Embeddings

Frameworks & Tools: PyTorch, Scikit-learn, Hugging Face, LangChain, LangGraph, RAGAS, FAISS, Chroma, FastAPI, Docker, Git, GitHub

Cloud: AWS (EC2, S3, RDS)

CERTIFICATIONS

AWS Certified AI Practitioner

AI/ML fundamentals, generative AI, and responsible AI on AWS.

Oracle Certified Data Science Professional

Cloud infrastructure and data science tooling for analytics workflows.

ServiceNow University – SmartBridge (AICTE)

8-module ServiceNow & Agentic AI internship – Flow Designer, ITSM workflows, ATF; CSA exam prep.